

# Identifying Research Methodologies in Computing Research Papers: Document-Level Extraction with a Long-Context LLM

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Report (6286 words, excluding tables, figures, references, and appendices)

## 1. Introduction (367/1000 words)

When computing researchers do a literature review, they often need to read many papers and find each paper's method, task, dataset, and evaluation metric. Reading many papers this way is slow and manual. I treat these four items as a methodology profile that a reader can extract automatically, to support the first pass of a literature review, not to replace reading the paper.

Consider "Attention Is All You Need" [D6]:

| Methodology role  | Component               |
|-------------------|-------------------------|
| Technical method  | Transformer             |
| Task              | machine translation     |
| Dataset           | WMT 2014 English-German |
| Evaluation metric | BLEU                    |

Table 1: Role-based methodology profile for "Attention Is All You Need" [D6].

A reader needing this summary currently has to read the paper and construct it themselves. A topic label answers "what is this paper about?"; a role-based profile answers "how was this research conducted?" Two papers on the same topic can use different methods, train on different datasets, and report different metrics, so a topic label alone does not answer that second question.

I use Template 12.1 from the Natural Language Processing (NLP) module: identifying research methodologies used in computing research papers. The code repository is publicly available at <https://github.com/sanemat/uol-fp>.

This motivation has not changed since the preliminary report. What changed is the extraction approach. proto2, my first working

prototype, classified every sentence in a paper into one of the four roles using zero-shot natural language inference (NLI), producing a list of candidate sentences per role rather than one answer. proto3, the current prototype, reframes the task as document-level extraction: given a paper, a long-context large language model (LLM) returns one answer per role, each backed by a section heading and a verbatim quote as evidence. I also implemented a decomposed variant of proto3 (four role-specific calls instead of one joint call) and a series of pilots aimed at Task, the weakest role.

The primary users are computing students doing literature reviews. Secondary users are early-stage researchers or supervisors who want a quick overview of a paper. The output is designed to be inspectable: a user can check the quoted evidence against the source paper.

## 1.1 Report structure

Chapter 2 reviews previous work on methodology extraction and zero-shot classification. Chapter 3 explains the design and evaluation plan, and Chapter 4 describes the implementation of proto2, proto3, and the Task experiments. Chapter 5 presents the results and their limitations. Chapter 6 concludes the report and discusses further work.

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## 2. Literature Review (1356/2500 words)

Chapter 1 showed a four-role profile for "Attention Is All You Need" [D6]. Figure 1 shows a fuller view of the same paper, including the design strategy and data generation method defined by Oates [14].

Methodology:

Design or strategy: design and creation + experiment

Data generation method: documents

Technical method: Transformer

Task: machine translation

Dataset: WMT machine translation datasets  
EvaluationMetric: BLEU score

Figure 1: Research Methodology from "Attention Is All You Need" [D6].

## 2.1 Defining Research Methodology

Research methodology in computing papers can be described using a structured vocabulary, but defining it is not the same as extracting it.

Oates [14] provides six research strategies (experiment, design and creation, survey, case study, action research, and ethnography) and four data generation methods (interviews, observations, questionnaires, and documents). His book defines the vocabulary that researchers use to describe their methodology in papers, so my project needs these concept names to identify what to extract. However, the six strategies were designed for human researchers to self-classify their own work — papers rarely contain the explicit phrase "this is an experiment". The vocabulary can be used to name what to look for, but it may not transfer directly to automatic extraction from text.

Pilkington & Pretorius [15] go further: they formalize the structure using UML (Unified Modeling Language) and ontology engineering, with the goal of "providing clear and unambiguous semantics" [15]. Key concepts are ResearchScheme, PhilosophicalWorldview, ResearchDesign, and ResearchMethod: a ResearchScheme is underpinned by one PhilosophicalWorldview and has one or more ResearchDesigns, and each ResearchDesign has one or more ResearchMethods.

A philosophical worldview is one of the key components in Pilkington & Pretorius [15], but it tends not to appear as an explicit phrase in completed research papers, so I exclude it from extraction. Research design (e.g. experiment vs. survey) [10] is similarly out of scope: it can be subjective, two readers can assign different labels to the same paper, and it is not the focus of this project's four roles.

Oates [14] gives concept names. Pilkington & Pretorius [15] give formal relationships between those concepts. My project uses vocabulary from Oates and formal structure from Pilkington & Pretorius. Both works are designed for human use; neither provides a system to extract methodology components automatically from text. The four roles used in this project (TechnicalMethod, Task, Dataset, and EvaluationMetric) draw more directly on Jain et al. [8], reviewed next.

## **2.2 Extracting Methodology from Papers**

Systems that extract methodology-like entities from papers exist [4, 5, 8], but the closest approaches are supervised and need labeled training data that I do not have.

Jain et al. [8] (SciREX) extract four entity types — Dataset, Metric, Task, and Method — that closely match the four roles in this project. They operate at the document level, arguing that "a significant amount of information can only be gleaned from analyzing the full document" [8] — relations may span sections, not just sentences. However, Jain et al. annotated 438 papers with a main annotator, and four PhD-level experts checked agreement on five documents (Cohen's  $\kappa$  95% for mention classification). The annotation builds on the Papers with Code corpus of 1,170 ML-conference articles, which covers only ML benchmarks. My project targets general computing papers (systems, algorithms, human-computer interaction (HCI), and ML research) where no comparable annotated dataset is available, so neither the corpus scope nor the annotation effort behind SciREX transfers directly.

Ma et al. [13] propose a metric-driven mechanism schema in which a mechanism is an (operation, effect, direction) triple linked to a task, and extract it from ACL abstracts using a query-guided sequence-to-sequence model, but their work is limited to the NLP domain and does not extend to general computing research.

Ghosh et al. [4, 5] use supervised transformer-based sequence labeling to extract methodology component names from AI research

papers, arguing that such names are large, fast-evolving, domain-specific, and context-dependent. Färber et al. [3] extract methods and datasets using domain-specific named entity recognition (NER) followed by usage classification, which distinguishes whether each mention is used by the authors or only mentioned. This "used vs. non-used" distinction is directly relevant to a noise problem this project also has: a sentence describing a prior method (for example, ELMo, cited in BERT's Introduction) can be misread as the paper's own method.

Both approaches need labeled training data that I do not have, and neither covers all four roles across general computing research: Ghosh et al. cover only TechnicalMethod, and Färber et al. only Method and Dataset. A zero-shot approach removes the need for annotated data, at the cost of domain adaptation, reviewed next.

## 2.3 Zero-shot Classification

Yin et al. [18] define zero-shot text classification as assigning a label to text without any task-specific training examples. They show that natural language inference (NLI) can classify text into many possible labels by turning the label into a hypothesis (for example, "this text is about sports") and asking a model whether the text entails it. I applied this approach directly in proto2, my first prototype: four methodology-role hypotheses, one per role, classified independently against each sentence in a paper, using a DeBERTa-v3-based zero-shot classification model [6].

| <b>aspect</b> | <b>labels</b>   | <b>interpretation</b>   | <b>example hypothesis (word)</b> |
|---------------|-----------------|-------------------------|----------------------------------|
| topic         | sports<br>etc.  | this text is about ?    | "?" = sports                     |
| emotion       | anger<br>etc.   | this text expresses ?   | "?" = anger                      |
| situation     | shelter<br>etc. | The people there need ? | "?" = shelter                    |

*Table 2 (adapted from Yin et al. [18]): example hypotheses for three general-domain task types.*

A domain mismatch risk exists: Yin et al. test on Yahoo News articles, emotion tweets, and crisis situation reports, and their NLI model is trained on MNLI (Multi-Genre Natural Language Inference; covers news, fiction, and telephone speech) — none of these are scientific papers, which tend to use dense technical vocabulary, passive constructions, and section-based structure. proto2 tested this risk directly by comparing hypothesis wordings on scientific text: a verbose TechnicalMethod hypothesis sent 244 of 258 BERT sentences to EvaluationMetric, a single poorly chosen hypothesis collapsing the label distribution, while short labels gave the best probe score and the most balanced distribution across four roles. Even with short labels, classification errors remained: on the Transformer paper, "Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task" was classified as Task rather than EvaluationMetric. Hypothesis design is therefore a real cost in zero-shot systems, and it did not fix the design-level problem in proto2's own results, covered next.

## **2.4 proto2's Own Findings as a Negative Result**

Sentence-level NLI classification produced too many candidate sentences to be usable — 151 TechnicalMethod sentences for the MapReduce paper alone — and its recall-only substring evaluation (18/24 across six papers) only checked whether a gold term appeared somewhere in the output, not whether the output itself was correct. With 100+ accepted sentences in some roles, a substring match is nearly certain to succeed somewhere in the list, which inflates the apparent recall without saying anything about precision. Excluding Related Work by heading, which describes other papers rather than the target paper's own methodology, cut BERT's TechnicalMethod count from 67 to 62 sentences; excluding the whole Introduction cut it further to 54, but also removed sentences that correctly described BERT's own method, so full exclusion traded recall for precision rather than solving the underlying problem. proto2 also had no mechanism to separate a paper's own method from one it cites: a sentence



describing ELMo in BERT's Introduction scored 0.87 as TechnicalMethod, even though ELMo is prior work, not BERT's own method. This sharpens Färber et al.'s [3] "used vs. non-used" gap (Section 2.2) with a concrete instance from my own data, and motivates proto3's document-level extraction with an explicit authorship rule (Chapter 3).

## 2.5 Document-Level and LLM-Based Structured Extraction

Jain et al. [8] argue that "a significant amount of information can only be gleaned from analyzing the full document" — a document-level information extraction (IE) claim. My own data supports this directly: Dataset and EvaluationMetric typically appear only in a paper's Experiment section, not the Abstract, so an extraction method effectively limited to a small set of sentences, as proto2's Introduction-heavy output tended to be, would miss them.

Dagdelen et al. [2] and Polak and Morgan [16] use LLMs to extract structured scientific information, including materials data. Ateia et al. [1] apply LLMs to extract information from scientific papers, the closest match in domain to this project. I adapt this approach to four methodology roles in computing papers, adding an authorship rule (Section 2.4) to distinguish the target paper's own work from cited methods, and a document-level context window (Chapter 3).

## 2.6 Synthesis

The reviewed studies provide the four-role schema and techniques for extracting information from scientific papers. However, their datasets and task definitions do not directly cover general computing research. I therefore tested schema-guided extraction without task-specific annotated training data.

| Source     | Contribution | Strength      | Limitation   | Relevance to this project |
|------------|--------------|---------------|--------------|---------------------------|
| Oates [14] | Methodology  | Clear concept | Designed for |                           |

| Source                      | Contribution                                                   | Strength                                                | Limitation                                         | Relevance to this project                                                                        |
|-----------------------------|----------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------|
|                             | vocabulary                                                     | names for methodology components                        | human use; not extraction-oriented                 | Motivates structured extraction                                                                  |
| Pilkington & Pretorius [15] | Formal methodology ontology                                    | Formal relationships between concepts                   | No extraction system or corpus                     | Supports treating methodology as a structured domain                                             |
| Jain et al. [8]             | Document-level extraction of Dataset, Metric, Task, and Method | All four roles; working system; document-level argument | 438 annotated ML papers; expert annotators         | Confirms four roles; motivates document-level extraction and both proto2 and proto3's approaches |
| Ghosh et al. [4, 5]         | TechnicalMethod extraction from AI papers                      | Methodology-specific sequence labeling                  | One role only; AI papers; supervised               | Shows difficulty of extracting method names                                                      |
| Färber et al. [3]           | Used vs. non-used methods and datasets                         | Handles authorship attribution                          | Method and Dataset only; labeled mentions required | Motivates proto3's authors'-own-work rule                                                        |
| Yin et al. [18]             | Zero-shot NLI text                                             | No task-specific                                        | Tested on general-                                 | Basis for proto2;                                                                                |

| Source                                                       | Contribution                                         | Strength                                                  | Limitation                                               | Relevance to this project                         |
|--------------------------------------------------------------|------------------------------------------------------|-----------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------|
|                                                              | classification                                       | training data needed                                      | domain text only; sentence-level                         | superseded by document-level extraction in proto3 |
| Dagdelen et al. [2]; Polak and Morgan [16]; Ateia et al. [1] | LLM-based structured extraction from scientific text | Document-level, schema-guided, no annotated corpus needed | Applied to other domains or to unstructured key concepts | Basis for proto3's schema-guided LLM extraction   |

Table 3: Key sources for this project.

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### 3. Design (1190/2000 words)

The system extracts research methodology from computing papers. The end-to-end input is a PDF, which a local GROBID [11] server converts to TEI XML before the notebook pipeline starts. The output is a role-based profile (Table 1, Chapter 1).

#### 3.1 Domain and Users

The domain and users are unchanged from the preliminary report. The domain is computing research papers, mainly systems, machine learning (ML), algorithms, and human-computer interaction (HCI). The primary users are computing students doing literature reviews; secondary users are early-stage researchers or supervisors who want a quick overview of a paper. proto3 changed the output quality — one checkable answer per role instead of a list of candidate sentences — not the target domain or audience.

| User need                             | System requirement          | Evaluation                                        |
|---------------------------------------|-----------------------------|---------------------------------------------------|
| Quickly understand a paper            | Show the four roles clearly | Output size (one answer per role, Chapter 4)      |
| Check the original paper              | Show evidence sentences     | Evidence correctness (manual review, Section 5.4) |
| Avoid missing important information   | High recall                 | Recall (Sections 5.2-5.3)                         |
| Avoid too much irrelevant information | Reduce noisy output         | Precision (Sections 5.2-5.3)                      |

Table 4: User needs mapped to system requirements and evaluation.

### 3.2 Design Justification

The core feature is Stage 2: one structured, evidence-backed answer per role, not a list of 14-160 candidate sentences (Section 2.4). proto2 classified each sentence independently with a fixed 0.5 threshold, which assumed that a sentence carries at most one role and that one threshold suits all roles. proto3 gives the model the whole document and asks for one answer per role, with a section heading and a verbatim quote as evidence, so a reader can check the answer without reading the paper. The authors'-own-work rule in the prompt targets proto2's authorship-attribution failure (the ELMo/BERT case, Section 2.4). The four-role JSON shape is enforced by `response_json_schema`, generated from Pydantic models rather than described in the prompt text, which removes an entire class of parsing and output-shape bugs (Chapter 4). A role can be null, so the model can report a role as absent instead of inventing one.

Extraction can be joint or decomposed. Variant A, the main pipeline, makes one call that returns all four roles; Jain et al.'s [8] document-level argument (Chapter 2) favours this design, since one context lets

the model link a method, dataset, and metric that appear together (e.g. "Transformer"/"WMT"/"BLEU"). Variant B makes four independent role-specific calls (Stage 2d). Khot et al. [9] show that decomposing a complex task into independently optimizable subtasks can beat a single joint few-shot prompt on several reasoning tasks, and the four roles ask for different judgments: primary method versus component, the problem actually solved, data actually used, and the metric actually reported. Both variants use the same RoleExtraction schema and the same scoring code, so they differ only in the number of calls and in one role-specific rule line. My hypothesis is that per-role accuracy under Variant B exceeds Variant A, which Section 5.5 tests. Variant C, a fifth call checking the four outputs for mutual consistency, is designed but not implemented (Chapter 6).

The second design question is whether every role should stay single-valued. AlexNet and ResNet both report top-1 and top-5 error rates, which the baseline squashes into one string, and BERT's gold EvaluationMetric label lists both "accuracy" and "F1" (Appendix A, Table A1), since the paper reports both. An informal NotebookLM cross-check, run independently on each paper without being told the schema was single-valued, produced similar multi-valued outputs for BERT, ResNet, and Transformer. BERT's single-valued Dataset answer is "SQuAD v1.1," while NotebookLM listed BooksCorpus, Wikipedia, GLUE, and SQuAD v1.1/v2.0 from the same source text. Forcing Dataset and EvaluationMetric into one string loses information.

Task shows the same pattern in one case: Variant B answered "sequence transduction" for Transformer where Variant A answered "machine translation", two defensible answers at different granularity (Section 5.5). Stage 2e therefore implements a multi-valued pilot for Task only (Chapter 4), followed by verification and reasoning-first variants. The main design keeps one answer per role. A multi-valued schema for Dataset and EvaluationMetric would need per-item evidence rather than one shared quote per list, and a ranked "primary first" order rather than a numeric confidence field, since this project's own measured non-determinism (Chapter 5) argues against a second, uncalibrated confidence axis. That schema remains a design proposal (Chapter 6).

### 3.3 Model Choice

A paper's cleaned full text is typically 4,000-20,000 tokens, which fits within the context window of several modern long-context LLMs without chunking:

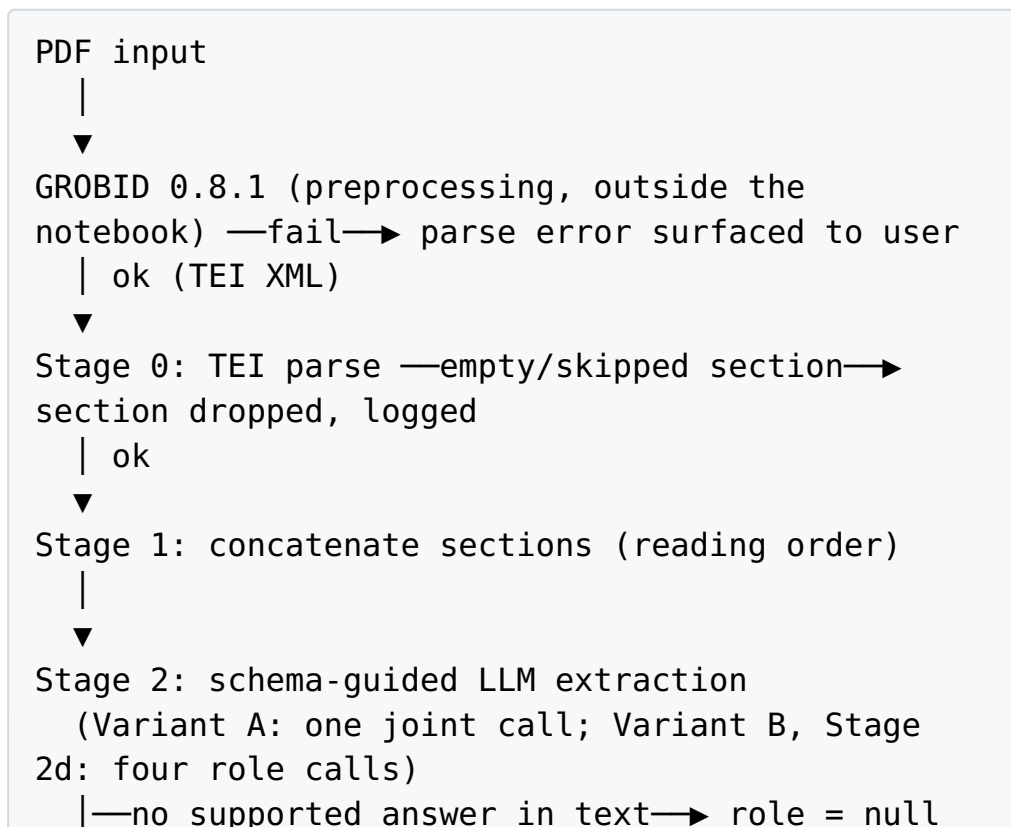
| Model        | Context   | Cost      |
|--------------|-----------|-----------|
| Gemini Flash | 1M tokens | cheap API |

Table 5: Long-context model.

I selected Gemini (gemini-3.5-flash, via the google-genai software development kit) for its long context window: 1M tokens exceeds the length of every paper in the corpus, so no chunking was needed. I did not compare other models. The API key comes from Colab's built-in secret manager (google.colab.userdata).

### 3.4 Overall Pipeline

The pipeline has one preprocessing step outside the notebook and five stages inside it (Figure 2).



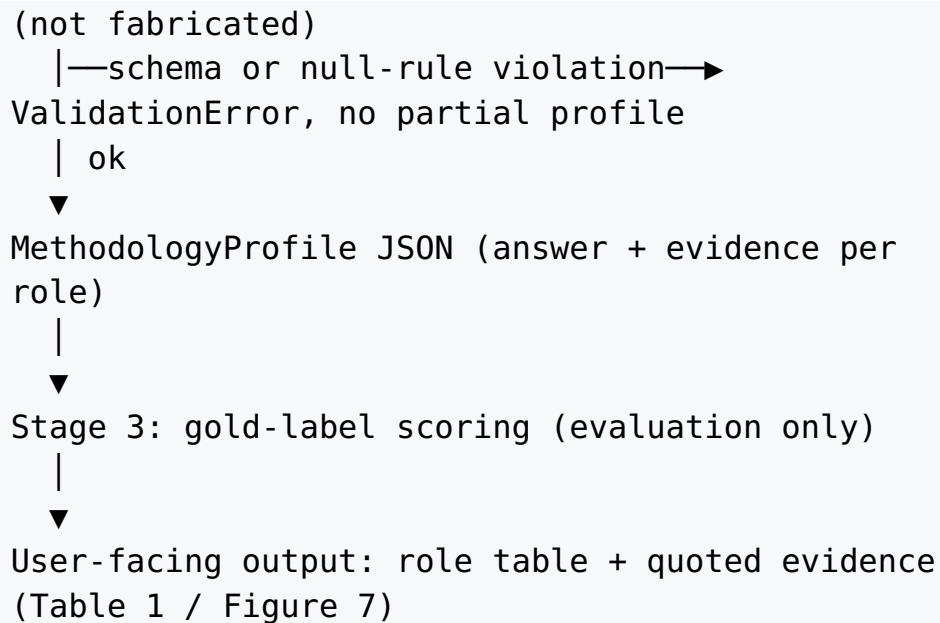


Figure 2: proto3 extraction pipeline with failure paths and the user-facing output step.

Compared with proto2's pipeline, there are two differences: there is no sentence splitting, and there is no per-sentence acceptance threshold. The LLM sees the (mostly) whole document and returns one decision per role directly, instead of a list of candidate sentences each scored independently (Figure 3).

```
proto2 (sentence level)
PDF → GROBID → TEI XML → section filter →
sentence split + clean
    → zero-shot NLI per sentence → threshold 0.5
→ 14-160 candidate sentences per role

proto3 (document level)
PDF → GROBID → TEI XML → concatenate sections
(reading order)
    → one LLM call + response_json_schema → one
answer + evidence per role
```

Figure 3: proto2 and proto3 pipelines compared. proto3 removes sentence splitting and the per-sentence acceptance threshold.

These two differences remove two weaknesses of proto2. First, proto2's NLI acceptance threshold of 0.5 needed a justification. proto3 has no such value, because extraction is no longer a per-sentence accept/reject decision. Second, proto2 assumed that a sentence describes one role at a time, and single-label sentence classification could only ever produce single-label results, whatever the underlying text actually contained. That circularity does not apply to proto3, since extraction is document-level; whether a *role* (not a sentence) should allow more than one answer is addressed in Section 3.2.

### 3.5 Evaluation Plan

proto2's plan treated a substring gold-label match with a 10-out-of-12 success threshold as sufficient, but a present-but-wrong answer cost nothing there. For proto3 I score gold-label match as a classification problem (true positive, false positive, false negative), report Precision, Recall, and F1 per role, and add Wilson 95% confidence intervals on Precision and Recall only — F1 is a harmonic mean, not a proportion, so a Wilson interval on it directly is not statistically meaningful, and a point-estimate F1 is appropriate at this sample size. I report both micro and macro averages, headlining macro, because the four roles are fixed, equally mandatory schema fields, not a frequency distribution.

I kept the sample at six papers: tightening the confidence intervals meaningfully would need roughly 30-40 gold-labeled papers per role, not the 6-10 reachable in the available time with no second annotator. This limits generalisability and remains a limitation of the evaluation. The plan also includes a logged variance study (repeat the pipeline several times rather than trust one run) and a single consolidated manual review pass covering plausibility, evidence support, authorship, and whether the quote appears in the source text.

I did not pool the five runs' true/false positive/negative counts into a single Wilson interval ( $n=30$  trials per role): the 30 trials are five repeats of the same six papers, not 30 independent observations, so



treating them as independent Bernoulli trials would overstate precision. The two measures stay separate: the  $n=6$  baseline Wilson interval for paper-level uncertainty, and the five-run F1 mean, minimum, maximum, and range for run-to-run non-determinism. Each answers a different question, and neither substitutes for the other.

For Task, the evaluation adds a comparison of Variant A and Variant B (one run each, scored with the unchanged `score_role`), and a sequence of pilots whose F1 is compared against the same gold labels (Section 5.5).

### 3.6 Iterations and Results

| Period               | Main task                                                                                                                       | Output                                                              | Status |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|--------|
| Before<br>29<br>June | Literature review, design, proto2 (sentence-level NLI)                                                                          | Preliminary Report                                                  | Done   |
| July                 | proto3 Stages 0-2 (document-level extraction); gold-label-match evaluation with Wilson confidence intervals on Precision/Recall | Baseline P/R/F1 table, 5-run F1 variance table (mean/min/max/range) | Done   |
| July-August          | Consolidated manual review pass; proto2 → proto3 fixed/not-fixed synthesis; figures for Implementation/Evaluation chapters      | Draft Report                                                        | Done   |
| Late August          | Variant B (decomposed extraction), scored against Variant A                                                                     | Table 13                                                            | Done   |
| Late August          | Task pilots (Stages 2e-2h) and LLM-judge rescoring (Stage 4)                                                                    | Table 14                                                            | Done   |

| Period      | Main task                               | Output           | Status      |
|-------------|-----------------------------------------|------------------|-------------|
| Late August | Variant C; Related Work ablation        | —                | Not run     |
| September   | Freeze experiments; final report; video | Final submission | In progress |

Table 6: Iteration summary.

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## 4. Implementation (1012/2500 words)

This chapter describes the implementation of proto3, including TEI parsing, schema-guided extraction, the decomposed variant, and the Task experiments. The source code and evaluation tools (`scoring.py`, `aggregate_runs.py`, `aggregate_variant_b.py`, with `pytest` tests) are available at <https://github.com/sanemat/uol-fp>.

### 4.1 Features Implemented

The prototype takes GROBID TEI XML and produces one JSON object containing an answer and evidence for each of the four roles (full example below). Parsing reuses proto2's approach. Every run can be scored against gold labels (Stage 3), and every answer can be checked against its quoted evidence.

### 4.2 Algorithms and Techniques

The end-to-end input is a PDF. A local GROBID 0.8.1 server converts it to TEI XML before the notebook runs (Appendix B), because GROBID keeps the section structure that Stages 0-1 need. This step is outside the notebook pipeline (`proto3/3pipeline.ipynb`), which takes the TEI XML as its input.

**Stage 0, parse the TEI XML.** `xml.etree.ElementTree` reads the file with the TEI namespace. The Abstract is taken from `tei:abstract`. Body sections are the `tei:div` elements under `tei:body`, with the heading from `tei:head` and the text from the `tei:p` children. Sections headed "references", "acknowledgements", or "acknowledgments" (case-insensitive) are skipped, as are sections with an empty body. The Abstract becomes the first section.

**Stage 1, concatenate.** Each section becomes `## {heading}` followed by its text, joined in reading order into one document. There is no sentence splitting, no chunking, and no per-sentence threshold, a direct response to `proto2`'s output-volume problem (Chapter 2). Dataset and EvaluationMetric often occur only in the Experiment section, so I keep the full document rather than an excerpt.

**Stage 2, schema-guided extraction.** The prompt states the four role definitions and three rules. The four-role JSON shape is not described in the prompt. It is passed as `response_json_schema`, generated from the MethodologyProfile Pydantic model, with `temperature=0` and `seed=0`, and the reply is parsed with `MethodologyProfile.model_validate_json`. `extra="forbid"` rejects unknown fields, and a `model_validator` rejects a role where only one of answer and evidence is null.

**Stage 2d, decomposed extraction (Variant B).** Four independent calls, one per role, each with a role-specific prompt and the smaller RoleExtraction schema. Each prompt names the failure that role is prone to: TechnicalMethod must be the primary method, not a component or prior work; Task must be the problem the paper actually solves; Dataset must be used for training or evaluation, not only mentioned; EvaluationMetric must report the paper's own results. The four results are assembled into the same `{role: answer}` shape that Stage 3 reads.

**Stages 2e-2h, Task pilots.** Stage 2e uses MultiValuedRoleExtraction, a list of answers at different granularities with the primary answer first. Stage 2f asks a second call to select one candidate from that list. Stages 2g and 2h use

ReasoningFirstRoleExtraction, in which a reasoning field precedes answer, following the "Thought of Structure" idea of Lu et al. [12] (reasoning before structured output) adapted to prompt time; 2g selects from Stage 2e's candidates and 2h extracts directly.

**Stage 3, evaluation.** `score_role` returns (tp, fp, fn, tn) per paper and role. A match is a normalised (lowercase, whitespace-collapsed) substring test in either direction. A both-present mismatch counts as one false positive and one false negative, and a null answer against a non-null gold label counts as a false negative only. `score_role_multi` accepts a match at any list position, and `precision_recall_f1` computes the metrics from the sums.

**Stage 4, LLM judge.** `score_role_judged` takes the match test as an injected function, so a Gemini semantic-equivalence check (returning a `SameTaskVerdict`, one boolean field) can replace the substring test while the tests use a stub and need no API call.

### 4.3 Code Explanation

Four details are the most technically interesting. First, the prompt states only what the JSON schema itself cannot express, and everything about the output shape lives on the Pydantic models (Figure 4):

```
class Evidence(BaseModel):
    section: str = Field(description="Exact
section heading containing the quote.")
    quote: str = Field(
description="One sentence quoted verbatim from
the paper, supporting answer."
)

class RoleExtraction(BaseModel):
    answer: str | None
    evidence: Evidence | None

    @model_validator(mode="after")
```

```

def answer_and_evidence_must_match(self) -> S
elf:
    if (self.answer is None) != (self.evidenc
e is None):
        raise ValueError(
            "answer and evidence must either
both be null or both be present"
        )
    return self

```

Figure 4: Evidence/RoleExtraction Pydantic models with the null-correlation validator (`proto3/src/uol\_fp/models.py`).

The validator enforces "no answer without evidence" in code. A JSON schema alone cannot express this rule, so a prompt instruction would be the only alternative, and it would be unenforced. The models live in `models.py`, and `sync_generated.py` (make sync-generated) generates the notebook block from that file, so the notebook and the tested module cannot drift apart.

Rules:

- Use the authors' own method, not methods cited from prior work.
- Return null when a role is not present in the paper.
- Evidence quotes must be copied verbatim from the paper, not paraphrased.

Figure 5: Prompt rules excerpt from `proto3/3pipeline.ipynb`, "Stage 2b — Prompt Template".

Second, the Gemini call uses `response_json_schema` together with `temperature=0` and `seed=0`, and parses the reply directly with `MethodologyProfile.model_validate_json(...)`, with no manual JSON-extraction step:

```

response = client.models.generate_content(
    model=MODEL_NAME,
    contents=prompt,

```

```

        config=types.GenerateContentConfig(
            temperature=0,
            seed=0,
            response_mime_type="application/json",
            response_json_schema=MethodologyProfile.model_json_schema(),
        ),
    )

    profile =
    MethodologyProfile.model_validate_json(response.text)

```

Figure 6: Gemini call and response parsing  
(`proto3/3pipeline.ipynb`, "Stage 2c — Call Gemini and Parse Response").

An earlier prompt version described the evidence field inconsistently. On "Attention Is All You Need", Gemini resolved the ambiguity by returning evidence as one flat string with the heading prepended, e.g. "## Introduction In this work we propose...", instead of the nested {section, quote} object. I first fixed this by rewriting the prompt; the schema now guarantees the nested shape regardless of prompt wording. `response_schema` and `response_json_schema` are not interchangeable. `response_schema=MethodologyProfile` fails with 400 `INVALID_ARGUMENT ... Unknown name "additional_properties"`, because it converts to Google's own Schema proto, which does not support `additionalProperties`, and Pydantic's `extra="forbid"` produces exactly that field. `response_json_schema` accepts a real JSON Schema dict instead, so `MethodologyProfile.model_json_schema()` is passed there.

Third, the Variant B prompts share one skeleton and the same two rules (null when absent, verbatim quotes), and only one rule line changes per role. The difference between Variant A and Variant B is therefore the decomposition plus that one line, which keeps the comparison controlled.

Fourth, the scoring semantics. Because matches is a substring test, the gold label "F1" matches "F1 score", while the gold Task "GLUE" does not match "language representation". Treating a wrong non-null answer as both a false positive and a false negative means precision penalises confident wrong answers and recall penalises abstention, which suits a tool whose output users are meant to trust. For code quality, pyright in strict mode and ruff report zero issues, and a pytest suite in proto3/tests/ covers scoring.py, the run aggregation, and the validator (make lint, make test). proto3/baseline.ipynb used to be a byte-identical duplicate of 3pipeline.ipynb, kept only because it had produced the six proto3/baseline/\*.json files; it has since been deleted, so there is no second notebook to keep in sync by hand.

## 4.4 Visual Representation

For "Attention Is All You Need" [D6], the full extraction output is:

```
{
  "TechnicalMethod": {
    "answer": "Transformer",
    "evidence": {
      "section": "Abstract",
      "quote": "We propose a new simple network
architecture, the Transformer, based solely on
attention mechanisms, dispensing with recurrence
and convolutions entirely."
    }
  },
  "Task": {
    "answer": "machine translation",
    "evidence": {
      "section": "Abstract",
      "quote": "Experiments on two machine
translation tasks show these models to be
superior in quality while being more
parallelizable and requiring significantly less
time to train."
    }
  }
},
```

```

"Dataset": {
  "answer": "WMT 2014 English-German",
  "evidence": {
    "section": "Training Data and Batching",
    "quote": "We trained on the standard WMT
2014 English-German dataset consisting of about
4.5 million sentence pairs."
  }
},
"EvaluationMetric": {
  "answer": "BLEU",
  "evidence": {
    "section": "Machine Translation",
    "quote":
"On the WMT 2014 English-to-German translation
task, the big transformer model (Transformer
(big) in Table 2) outperforms the best previously
reported models (including ensembles) by more
than 2.0 BLEU, establishing a new state-of-the-
art BLEU score of 28.4."
  }
}
}

```

Figure 7: Full extraction output for "Attention Is All You Need" [D6] (`proto3/baseline/transformer.json`).

Figure 8 shows the Stage 2c cell with the raw Gemini call and its parsed JSON output, so the extraction is visible running directly.



```

[12] ✓ 9%
response = client.models.generate_content(
    model=MODEL_NAME,
    contents=prompt,
    config=types.GenerateContentConfig(
        temperature=0,
        seed=0,
        response_mime_type="application/json",
        response_json_schema=MethodologyProfile.model_json_schema(),
    ),

    profile = MethodologyProfile.model_validate_json(response.text)

    for role in ROLES:
        print(f"{role}: {getattr(profile, role)}")

    print()
    print(profile.model_dump_json(indent=2))

... TechnicalMethod: answer='Transformer' evidence=Evidence(section='Abstract', quote='We propose a new simple network
Task: answer='machine translation' evidence=Evidence(section='Abstract', quote='Experiments on two machine transla
Dataset: answer='WMT 2014 English-German dataset' evidence=Evidence(section='Training Data and Batching', quote='W
EvaluationMetric: answer='BLEU' evidence=Evidence(section='Machine Translation', quote='On the WMT 2014 English-to

{
  "TechnicalMethod": {
    "answer": "Transformer",
    "evidence": {
      "section": "Abstract",
      "quote": "We propose a new simple network architecture, the Transformer, based solely on attention mechanism"
    }
  },
  "Task": {
    "answer": "machine translation",
    "evidence": {
      "section": "Abstract",
      "quote": "Experiments on two machine translation tasks show these models to be superior in quality while bei
    }
  },
  "Dataset": {
    "answer": "WMT 2014 English-German dataset",
    "evidence": {
      "section": "Training Data and Batching",
      "quote": "We trained on the standard WMT 2014 English-German dataset consisting of about 4.5 million sentenc
    }
  },
  "EvaluationMetric": {
    "answer": "BLEU",
    "evidence": {
      "section": "Machine Translation",

```

Figure 8: proto3 Stage 2c cell and output (screenshot).

Table 8 shows the contrast with proto2 numerically for the Transformer paper: proto2's accepted-sentence counts per role versus proto3's one answer per role.

| Paper               | proto2<br>Technical<br>Method | proto2<br>Task | proto2<br>Dataset | proto2<br>EvaluationMetric | proto3                             |
|---------------------|-------------------------------|----------------|-------------------|----------------------------|------------------------------------|
| Transformer<br>[D6] | 14<br>sentences               | 0<br>sentences | 0<br>sentences    | 160<br>sentences           | 1 answer +<br>evidence<br>per role |

Table 8: proto2 sentence-count output vs proto3 answer-and-evidence output, Transformer paper.

Table 9 compares the Transformer output of Variant A and Variant B. Only Task differs, and the Variant B answer comes from a Conclusion sentence: "In this work, we presented the Transformer, the first sequence transduction model based entirely on attention..."

| Role              | Variant A                       | Variant B                       |
|-------------------|---------------------------------|---------------------------------|
| Technical Method  | Transformer                     | Transformer                     |
| Task              | machine translation             | sequence transduction           |
| Dataset           | WMT 2014 English-German dataset | WMT 2014 English-German dataset |
| Evaluation Metric | BLEU                            | BLEU                            |

Table 9: Variant A (proto3/results/run1) and Variant B (proto3/results\_b) answers for the Transformer paper.

---

## 5. Evaluation (1930/2500 words)

### 5.1 Evaluation Method

I evaluate gold-label match as a classification problem — Precision, Recall, and F1 per role, with Wilson 95% confidence intervals on Precision and Recall — backed by a logged 5-run variance study and a consolidated manual review pass covering plausibility, evidence support, authorship, and whether the quote appears in the source text, all in a single read. Variant B and the Task pilots are scored against the same gold labels.

Unlike proto2's recall-only substring check, Precision/Recall/F1 (P/R/F1) penalises a present-but-wrong answer.

The manual review includes the verbatim-quote check because a verbatim match shows only that the LLM copied text, not that the text is good evidence (a real quote from a Related Work sentence can still be the wrong evidence).

## 5.2 Gold-Label-Match Results

Scoring the frozen baseline (proto3/baseline/\*.json) against gold labels (Appendix A, Table A1) across all six papers gives:

| <b>Role</b>      | <b>P</b> | <b>R</b> | <b>F1</b> |
|------------------|----------|----------|-----------|
| TechnicalMethod  | 0.83     | 0.83     | 0.83      |
| Task             | 0.33     | 0.33     | 0.33      |
| Dataset          | 0.80     | 0.67     | 0.73      |
| EvaluationMetric | 0.80     | 0.67     | 0.73      |
| Micro            | 0.68     | 0.62     | 0.65      |
| Macro            | —        | —        | 0.655     |

Table 10: Baseline gold-label-match results, all six papers.

Macro is the headline score (Section 3.5). The two averages are close (0.65 vs 0.655) because every role has  $n=6$  in this dataset.

Wilson 95% confidence intervals show the effect of the sample size: TechnicalMethod recall 0.83 gives a confidence interval of [0.44, 0.97]; Task recall 0.33 gives [0.10, 0.70]. These substantially overlap, so I do not claim TechnicalMethod is reliably "solved" while Task is reliably "broken" at this sample size. I report F1 as a point estimate, without a Wilson interval (Section 3.5). One gold label carries a specific evaluator-influence caveat: AlexNet's TechnicalMethod gold label was changed from "AlexNet" to "convolutional" after running the pipeline and inspecting its output, since the 2012 paper predates the name "AlexNet" and never uses it. Adjusting a gold label after seeing model output limits how far this result generalises, and it is one instance of a broader single-annotator problem: I wrote both the gold labels and, later, the answers checked against them (Section 5.4).

The screenshot shows a Google Colab notebook interface. The top bar includes a search bar, tabs for 'Commands', 'Code', 'Text', and 'Run all', and a 'Save in GitHub to keep changes' button. The left sidebar contains icons for file management and a 'Variables' panel. The main area displays a Python script with the following code:

```
return totals

def print_score_table(totals: dict[str, tuple[int, int, int, int]], title: str) -> None:
    print(title)
    print(
        f"{'Role':<18} {'TP':>3} {'FP':>3} {'FN':>3} {'TN':>3}"
        f" {'P':>5.2f} {'R':>5.2f} {'F1':>5.2f}"
    )
    overall = (0, 0, 0, 0)
    for role in ROLES:
        tp, fp, fn, tn = totals[role]
        p, r, f1 = precision_recall_f1(tp, fp, fn)
        print(
            f"{'role':<18} {'tp':>3} {'fp':>3} {'fn':>3} {'tn':>3}"
            f" {'p':>5.2f} {'r':>5.2f} {'f1':>5.2f}"
        )
        overall = tuple(a + b for a, b in zip(overall, (tp, fp, fn, tn)))
    p, r, f1 = precision_recall_f1(overall[0], overall[1], overall[2])
    print(
        f"{'Overall':<18} {'overall[0]:>3} {'overall[1]:>3} {'overall[2]:>3}"
        f" {'overall[3]:>3}"
        f" {'p':>5.2f} {'r':>5.2f} {'f1':>5.2f}"
    )
    print()

BASELINE_TOTALS = aggregate_scores(BASELINE_ANSWERS)
print_score_table(BASELINE_TOTALS, "Baseline (frozen proto3/baseline/*.json) vs gold:")
```

The output of the script is a table showing the performance metrics for different roles:

| Role             | TP | FP | FN | TN | P    | R    | F1   |
|------------------|----|----|----|----|------|------|------|
| TechnicalMethod  | 5  | 1  | 1  | 0  | 0.83 | 0.83 | 0.83 |
| Task             | 2  | 4  | 4  | 0  | 0.33 | 0.33 | 0.33 |
| Dataset          | 4  | 1  | 2  | 0  | 0.80 | 0.67 | 0.73 |
| EvaluationMetric | 4  | 1  | 2  | 0  | 0.80 | 0.67 | 0.73 |
| Overall          | 15 | 7  | 9  | 0  | 0.68 | 0.62 | 0.65 |

The bottom part of the notebook shows the code for comparing the pipeline answers with the gold answers:

```
pipeline_answers = {role: getattr(profile, role).answer for role in ROLES}
gold_answers = GOLD_LABELS[PAPER_SLUG]
baseline_answers = BASELINE_ANSWERS[PAPER_SLUG]

print(f"{'Role':<18} {'Gold':<25} {'Baseline':<25} {'Pipeline':<25}")
for role in ROLES:
    gold = gold_answers.get(role) or "(none)"
    baseline = baseline_answers.get(role) or "(none)"
    pipeline = pipeline_answers.get(role) or "(none)"
    print(f"{'role':<18} {'gold':<25} {'baseline':<25} {'pipeline':<25}")
print()

print_score_table(
```

The bottom status bar shows 'Variables', 'Terminal', a blue star icon, '9:32 AM', and 'Python 3'.

Figure 9: Baseline P/R/F1 scoring output  
(`proto3/3pipeline.ipynb`, Stage 3).

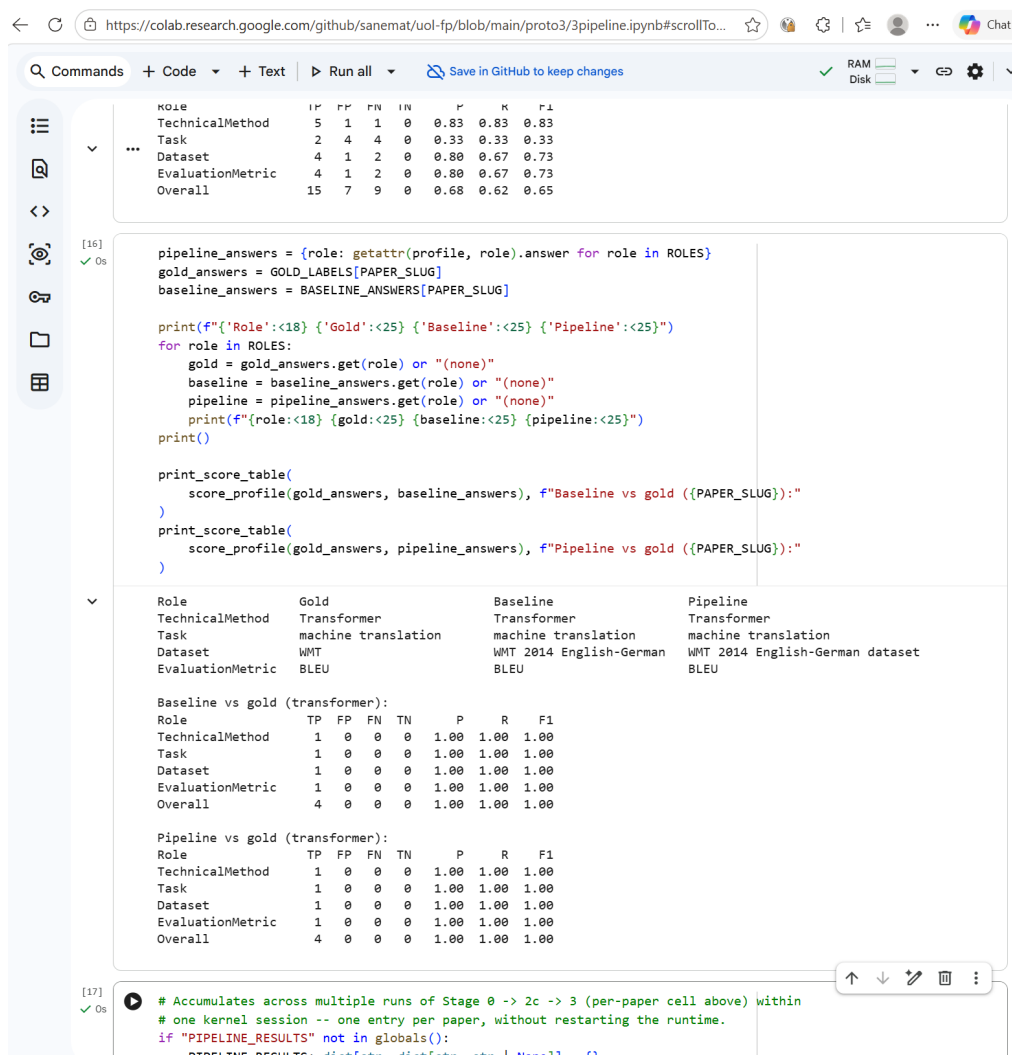


Figure 10: Per-paper gold-label scoring for the Transformer paper, baseline vs. pipeline (`proto3/3pipeline.ipynb`, Stage 3).

### 5.3 Variance Study

I logged five full pipeline runs to `proto3/results/run{1..5}/*.json` and aggregated them with `proto3/aggregate_runs.py`. Per-role F1 across the five runs:

| Role            | F1 mean | F1 min | F1 max | F1 range |
|-----------------|---------|--------|--------|----------|
| TechnicalMethod | 0.83    | 0.83   | 0.83   | 0.00     |
| Task            | 0.33    | 0.33   | 0.33   | 0.00     |
| Dataset         | 0.91    | 0.91   | 0.91   | 0.00     |

| Role             | F1 mean | F1 min | F1 max | F1 range |
|------------------|---------|--------|--------|----------|
| EvaluationMetric | 0.57    | 0.33   | 0.67   | 0.33     |

Table 11: Per-role F1 across 5 real pipeline runs (proto3/results/aggregate.json).

Three of four roles were perfectly stable across five real repetitions, which is stronger evidence than an earlier two-run anecdote in which both Dataset and EvaluationMetric had moved. At  $n=5$ , only EvaluationMetric varied (F1 ranged 0.33-0.67 across runs, with unchanged code, temperature=0, and seed=0), which narrows the non-determinism finding. The frozen baseline behind Table 10 is not a like-for-like sixth run alongside these five: it was generated before temperature=0 and seed=0 were added to the Gemini call, so only the five logged runs share identical settings. Across the five runs, TechnicalMethod's F1 (0.83) exceeded Task's F1 (0.33) every time.

An informal cross-check with Google NotebookLM, run independently on each paper, found stable agreement on TechnicalMethod across all six papers — exact or near-exact matches including "Google", "BERT", "Transformer", and "MapReduce". This is independent corroboration that TechnicalMethod is the strongest role.

The five runs are not pooled into a Wilson interval (Section 3.5). The  $n=6$  intervals in Section 5.2 cover paper-level uncertainty, and Table 11 covers run-to-run variation on the same papers.

MapReduce's Task slot (gold "distributed", system answer "automatic parallelization and distribution of large-scale computations") fails the substring-match rule despite being arguably correct. Task's low F1 is therefore partly an artifact of the measurement instrument and not purely a model failure.

MapReduce's Dataset slot (gold "TeraSort") answered null in every one of the five runs, a genuine recall miss, since NotebookLM independently found the dataset description (two roughly 1 terabyte grep/sort benchmarks) in the same source text. Pagerank's EvaluationMetric slot (gold "quality") answered "precision" in

every one of the five runs. It is scored wrong by substring match, but the paper's text supports both terms, and NotebookLM's independent extraction also names precision.

## 5.4 Manual Review

The review template stages each paper's answer, section, and quote next to four judgment columns (plausible? evidence supports? authors' own work? quote in source?). The template also staged two open questions before scoring. BERT's Task quote cites prior work by name, so the review had to decide whether pre-training is BERT's own contribution or only motivation. ResNet's Task quote cites bracketed prior work for "a series of breakthroughs," which raises the same question of whether the sentence establishes the paper's own task or credits others. All 24 slots are scored. Two are null (Pagerank/EvaluationMetric, MapReduce/Dataset), judged separately below; across the other 22 scored slots:

| Check              | Failures (of 22) | Slots                                                                                    |
|--------------------|------------------|------------------------------------------------------------------------------------------|
| Plausible?         | 2                | Pagerank/TechnicalMethod, Pagerank/Task                                                  |
| Evidence supports? | 5                | Pagerank/TechnicalMethod, Pagerank/Task, AlexNet/Task, BERT/Task, BERT/Dataset           |
| Authors' own work? | 6                | Pagerank/Task, AlexNet/Task, BERT/Task, BERT/Dataset, BERT/EvaluationMetric, ResNet/Task |
| Quote in source?   | 1                | ResNet/Task                                                                              |

Table 12: Manual review failure counts by check type, 22 scored slots.

Six slots pass the quote-in-source check but still fail on evidence-support or authorship — a real, verbatim quote that is still the *wrong* evidence, distinct from a fabricated quote: Pagerank/TechnicalMethod ("Google is the proposed system, not the technical method"),

Pagerank/Task ("information retrieval is the broader problem domain, not the task performed by the proposed system"), AlexNet/Task ("the quote does not directly prove that this is AlexNet's own task"), BERT/Task (the quoted sentence "describes prior work rather than BERT's own contribution"), BERT/Dataset ("valid dataset, but one of several"), and BERT/EvaluationMetric ("F1 is one of several evaluation metrics used across BERT's downstream tasks"). ResNet's Task answers the second open question above and is the one slot where quote-in-source and authorship fail together: the quote "cites bracketed prior work [21, 49, 39] for the breakthroughs," crediting prior work rather than establishing the paper's own task.

On the two null slots: MapReduce's Dataset is a real miss, not a genuine absence — the paper "uses large benchmark datasets for grep and sort experiments (about 1 TB each)," which NotebookLM independently found in the source text. Pagerank's EvaluationMetric leans toward a gold-label problem: precision "is discussed as an important search-quality criterion, but it is not actually measured or reported as an experimental metric" — gold "quality" is likely too vague or mislabeled.

BERT's Dataset and EvaluationMetric failures confirm the multi-valued-roles finding from Chapter 3: choosing only one of several correct answers can be technically correct but still gives an incomplete picture. Some gold answers may also be too simple or use the wrong category, so evaluating against gold labels alone can give a misleading result. The review has the same single-annotator bias as the gold labels it checks against: I wrote both the gold labels and, later, the answers checked against them, so I do not present this review as more objective than the gold labels themselves.

## **5.5 Decomposed Extraction and the Task Pilots**

Variant B (four role-specific calls) ran once on all six papers and was scored against run 1 of Variant A, so both columns are single runs:



| Role             | A F1 | B F1 | $\Delta F1$ |
|------------------|------|------|-------------|
| TechnicalMethod  | 0.83 | 0.83 | +0.00       |
| Task             | 0.33 | 0.33 | +0.00       |
| Dataset          | 0.91 | 1.00 | +0.09       |
| EvaluationMetric | 0.67 | 0.67 | +0.00       |

Table 13: Variant A (results/run1) vs Variant B (results\_b), F1 per role, one run each.

Only Dataset moved (+0.09), and its prompt was not written against a known Dataset failure. Task's F1 stayed at 0.33, but the two matching papers differ. For Pagerank, Variant B replaced an Introduction sentence about the Web's challenges for information retrieval (rated wrong in the manual review) with "web search", taken from the Abstract's statement of what the paper addresses, and the substring test now matches. For Transformer, Variant B answered "sequence transduction" from a Conclusion sentence describing the architecture, where Variant A had answered "machine translation" from the Abstract's evaluated task (Table 9). The role-specific rule separates the paper's own claim from prior work, but it cannot prefer the evaluated benchmark over a broader self-description, and the two changes cancel. Variant C only reconciles Variant B's four outputs with each other, so it could not resolve this ambiguity, and I did not run it.

Five further mechanisms targeted Task on the same six papers, each run once (Table 14). Stage 2e asked for every valid Task description, most specific first. Stage 2f asked a second call to select one candidate. Stage 4 replaced the substring match with a Gemini judge on Variant A's four mismatches. Stages 2g and 2h asked the model to write its reasoning before its answer, either to select from Stage 2e's candidates (2g) or to extract directly (2h), adapting Lu et al. [12].

| Mechanism         | P    | R    | F1   | Note             |
|-------------------|------|------|------|------------------|
| Variant A (joint) | 0.33 | 0.33 | 0.33 | 2 TP, 4 FP, 4 FN |

| <b>Mechanism</b>                           | <b>P</b> | <b>R</b> | <b>F1</b>                            | <b>Note</b>                                              |
|--------------------------------------------|----------|----------|--------------------------------------|----------------------------------------------------------|
| Variant B<br>(decomposed)                  | 0.<br>33 | 0.<br>33 | 0.33                                 | different two papers<br>match                            |
| Stage 2e, any-<br>match                    | 0.<br>67 | 0.<br>67 | 0.67                                 | match anywhere in a<br>list of 2-7 items                 |
| Stage 2e,<br>primary answer<br>only        | 0.<br>17 | 0.<br>17 | 0.17                                 | only BERT's first item<br>matched                        |
| Stage 2f,<br>verification                  | 0.<br>33 | 0.<br>33 | 0.33                                 | BERT's correct answer<br>replaced                        |
| Stage 4, LLM<br>judge                      | —        | —        | 0.67 (about<br>0.50 after<br>review) | one of two SAME<br>verdicts contradicts<br>manual review |
| Stage 2g,<br>reasoning-first<br>selection  | 0.<br>50 | 0.<br>33 | 0.40                                 | gain from returning<br>null on BERT and<br>MapReduce     |
| Stage 2h,<br>reasoning-first<br>extraction | 0.<br>33 | 0.<br>33 | 0.33                                 | same hits as Variant A                                   |

Table 14: Task F1 under each mechanism, six papers, one run each (proto3/results\_\*).

The any-match score doubles F1 but is bounded only by the length of the candidate list. Of its four hits, only BERT's was the model's first choice; the others were listed second to fourth in lists of 2-5 items. Stage 2f and Stage 2g both removed BERT's answer "GLUE benchmark" (correct as Stage 2e's first item), replacing it with a wrong answer and with null respectively. Stage 2g's 0.40 comes entirely from abstention, since a null against a non-null gold label counts as a false negative only. The Stage 2g reasoning traces evaluate each candidate against an explicit granularity test, so the model reasons coherently and still lands on a different phrasing from the gold label or declines to answer.

The Stage 4 judge returned SAME for MapReduce's "automatic parallelization and distribution of large-scale computations" against "distributed", which agrees with the earlier reading in Section 5.3. It also returned SAME for Pagerank's "information retrieval" against "web search", which contradicts the manual review of the same answer (Section 5.4). The judge and the extractor are the same model family, and with four judged pairs I could not establish whether its verdicts can be trusted in general. A lexical alternative also fails on these pairs: Jaccard similarity between the gold label and the answer is 0.00-0.25, far below the 0.8 threshold that Ateia et al. [1] use.

These results agree with published findings. Tam et al. [17] report that restricting LLM output to a format can lower performance, which is compatible with the gap between Stage 2e's any-match (0.67) and primary-only (0.17) scores. Huang et al. [7] report that LLMs cannot reliably correct their own reasoning without external feedback, which is compatible with the regressions in Stages 2f, 2g, and 4. Ateia et al. [1] score free-text extraction targets with BERTScore F1 [19] and report a semantic F1 near 0.90 on those targets, while exact categorical extraction stays below 0.25. The two scores cover different target types, but they show that a lenient semantic score and a strict match can differ widely. Task's residual F1 may reflect the single-gold-phrasing substring metric as much as the model. A deterministic semantic-similarity rescoring is planned but not run; it would need one threshold that accepts the MapReduce pair and rejects the Pagerank pair.

Two limitations apply to the Task line. The Stage 2f prompt was revised while inspecting the Transformer failure, and its first version used two of the project's own gold Task labels as examples and was discarded before any data collection. Every mechanism is a single run on six papers, so differences of one paper (0.17 in F1) are within the run-to-run variation seen for EvaluationMetric in Table 11.

I did not run the Related Work ablation, which would exclude the Related Work section before extraction. It does not address Task, the weakest role, and the core claims do not depend on it, so it stays deferred to further work (Chapter 6).

## 5.6 Critical Evaluation

Mapping proto2's three named failure modes, and the Task weakness found later, onto what the evaluation measured across the whole project:

| Failure mode                                                                     | Status                                   | Evidence                                                                                                                                                   |
|----------------------------------------------------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Output volume (151 TechnicalMethod sentences for MapReduce)                      | Fixed by design                          | One answer per role, every paper, by construction of Stage 2's schema-guided extraction                                                                    |
| No authorship-attribution mechanism (ELMo scored 0.87 as BERT's TechnicalMethod) | Addressed by design, not reliably solved | The authors'-own-work rule targets this directly; the manual review found 6 of 22 scored slots still fail authorship, concentrated in Task (4 of 6 papers) |
| Recall-only evaluation (10/12, then 18/24 substring match)                       | Fixed                                    | Precision/Recall/F1 per role, Wilson confidence intervals on Precision/Recall, a 5-run variance study                                                      |
| Task F1 of 0.33                                                                  | Not fixed                                | Six mechanisms (Table 14) did not raise it above 0.33 under a check I could trust                                                                          |

Table 15: Failure modes across proto2, proto3, and the Task experiments.

The remaining problems are mainly semantic rather than structural. The authorship failures (6 of 22 scored slots) show that an explicit prompt rule cannot reliably distinguish a paper's own contribution from prior work. Task extraction also remains hard to evaluate. It stayed at F1 0.33 across five repeated baseline runs, and none of the additional mechanisms showed a reliable improvement: higher scores

depended on lenient matching or on abstention. The single gold-label phrasing can also reject defensible alternatives.

These findings are limited by the six-paper corpus, single-annotator gold labels and review, and single-run comparisons for Variant B and the Task experiments. AlexNet's gold label was changed after I inspected the output (Section 5.2). No user study was conducted, so the practical value of the extracted profiles for a first-pass literature review remains untested.

---

## **6. Conclusion (431/1000 words)**

### **6.1 Summary**

This project developed a document-level LLM pipeline to extract four methodology roles (technical method, task, dataset, and evaluation metric) from computing papers. proto3 replaces proto2's sentence-level classification with one evidence-backed answer per role.

TechnicalMethod and Dataset reach F1 0.83 and 0.91 across five repeated runs, but Task remains at 0.33. The Task experiments exposed two related problems: a single answer cannot always represent a paper's task, and substring matching can reject reasonable answers that differ from the gold label.

response\_json\_schema solved schema conformance, so the remaining difficulty lies in defining the four roles consistently across paper types. MapReduce and Google Search fit them poorly because the roles derive from ML-benchmark structure.

The prototype produces more concise and inspectable output than proto2, but the manual review found continuing problems with evidence and authorship attribution. Its usefulness for literature reviews has not been tested with users.

## 6.2 Further Work

The next experiments follow from the results above.

- **Task rescoring.** Stage 5 would rescore all Task variants with a deterministic semantic similarity (BERTScore [19] or embedding cosine) and test whether one threshold accepts the MapReduce pair and rejects the Pagerank pair. If none does, the semantic metric has the same leniency problem as the LLM judge.
- **Variant C.** A fifth call would check Variant B's four outputs against their evidence. Variant B's result gives little reason to expect a gain on Task.
- **Multi-valued roles.** A MultiRoleExtraction type with per-item evidence and a ranked list capped at three items would allow several Dataset and EvaluationMetric answers. It requires re-annotating four gold cells (BERT and Transformer Dataset, AlexNet and ResNet EvaluationMetric) and a parallel `score_role_multi` function.
- **Related Work ablation.** Excluding Related Work before extraction would test whether it reduces the authorship failures found in the manual review.
- **Larger and broader corpus.** About 30-40 gold-labeled papers per role, a second annotator (allowing an inter-annotator-agreement study), and non-ML papers such as systems and HCI would test whether the four-role schema generalises.
- **User study.** A study with computing students would measure whether the profiles support the first pass of a literature review.

## 6.3 Broader Theme

`response_json_schema` constrains the output format, and Pydantic validates the returned data. Neither guarantees that the extracted information is correct: an answer can be well-formed and still wrong, as the manual review in Chapter 5 shows. The Task experiments add a third case, where an answer is well-formed, well-reasoned, and marked wrong by the metric because it uses a different phrasing from the gold label. The distinction between schema conformance and

semantic correctness applies to LLM-based structured extraction generally.

---

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## Appendix A — Extended proto3 Output and Background Data (All 6 Papers)

Chapter 4 shows the full extraction output for the Transformer paper (Figure 7). This appendix gives the same output for the remaining five papers, all from `proto3/baseline/*.json`, plus the gold labels and proto2 background counts used throughout Chapters 4–5.

```
{
  "TechnicalMethod": {
    "answer": "deep convolutional neural network",
    "evidence": {
      "section": "Abstract",
      "quote": "We trained a large, deep convolutional
neural network to classify the 1.2 million high-resolution
images in the ImageNet LSVRC-2010 contest into the 1000
different classes."
    }
  },
  "Task": {
    "answer": "object recognition",
    "evidence": {
      "section": "Introduction",
      "quote": "Current approaches to object recognition
make essential use of machine learning methods."
    }
  },
  "Dataset": {
    "answer": "ImageNet",
    "evidence": {
      "section": "The Dataset",
      "quote": "ImageNet is a dataset of over 15 million
labeled high-resolution images belonging to roughly 22,000
categories."
    }
  }
}
```

```

    }
  },
  "EvaluationMetric": {
    "answer": "top-1 and top-5 error rates",
    "evidence": {
      "section": "The Dataset",
      "quote":
        "On ImageNet, it is customary to report two error rates:
        top-1 and top-5, where the top-5 error rate is the
        fraction of test images for which the correct label is not
        among the five labels considered most probable by the
        model."
    }
  }
}

```

Figure A1: Full extraction output for AlexNet [D5].

```

{
  "TechnicalMethod": {
    "answer": "BERT",
    "evidence": {
      "section": "Abstract",
      "quote":
        "We introduce a new language representation model called
        BERT, which stands for Bidirectional Encoder
        Representations from Transformers."
    }
  },
  "Task": {
    "answer": "Language model pre-training",
    "evidence": {
      "section": "Introduction",
      "quote":
        "Language model pre-training has been shown to be
        effective for improving many natural language processing
        tasks (Dai and Le, 2015; Peters et al., 2018a; Radford et
        al., 2018; Howard and Ruder, 2018) ."
    }
  },
  "Dataset": {

```

```

    "answer": "SQuAD v1.1",
    "evidence": {
      "section": "SQuAD v1.1",
      "quote": "The Stanford Question Answering Dataset (SQuAD v1.1) is a collection of 100k crowdsourced question/answer pairs (Rajpurkar et al., 2016) ."
    }
  },
  "EvaluationMetric": {
    "answer": "F1 score",
    "evidence": {
      "section": "SQuAD v1.1",
      "quote": "Our single BERT model outperforms the top ensemble system in terms of F1 score."
    }
  }
}

```

Figure A2: Full extraction output for BERT [D3].

```

{
  "TechnicalMethod": {
    "answer": "MapReduce",
    "evidence": {
      "section": "Abstract",
      "quote": "MapReduce is a programming model and an associated implementation for processing and generating large data sets."
    }
  },
  "Task": {
    "answer": "automatic parallelization and distribution of large-scale computations",
    "evidence": {
      "section": "Introduction",
      "quote": "The major contributions of this work are a simple and powerful interface that enables automatic parallelization and distribution of large-scale computations, combined with an implementation of this interface that achieves high performance on large clusters of commodity PCs."
    }
  }
}

```

```

    }
  },
  "Dataset": {
    "answer": null,
    "evidence": null
  },
  "EvaluationMetric": {
    "answer": "elapsed time",
    "evidence": {
      "section": "Effect of Backup Tasks",
      "quote":
"The entire computation takes 1283 seconds, an increase of
44% in elapsed time."
    }
  }
}

```

Figure A3: Full extraction output for MapReduce [D2].

```

{
  "TechnicalMethod": {
    "answer": "Google",
    "evidence": {
      "section": "Abstract",
      "quote": "In this paper, we present Google, a
prototype of a large-scale search engine which makes heavy
use of the structure present in hypertext."
    }
  },
  "Task": {
    "answer": "information retrieval",
    "evidence": {
      "section": "Introduction",
      "quote": "The Web creates new challenges for
information retrieval."
    }
  },
  "Dataset": {
    "answer": "24 million pages",
    "evidence": {
      "section": "Anchor-test",

```

```

    "quote": "In our current crawl of 24 million pages.
we had over 259 million anchors which we indexed."
  },
  "EvaluationMetric": {
    "answer": null,
    "evidence": null
  }
}

```

Figure A4: Full extraction output for Google Search (PageRank) [D1].

```

{
  "TechnicalMethod": {
    "answer": "deep residual learning framework",
    "evidence": {
      "section": "Introduction",
      "quote": "In this paper, we address the degradation
problem by introducing a deep residual learning
framework."
    }
  },
  "Task": {
    "answer": "image classification",
    "evidence": {
      "section": "Introduction",
      "quote": "Deep convolutional neural networks [22,
21] have led to a series of breakthroughs for image
classification [21, 49, 39].\"
    }
  },
  "Dataset": {
    "answer": "ImageNet 2012 classification dataset",
    "evidence": {
      "section": "ImageNet Classification",
      "quote": "We evaluate our method on the ImageNet
2012 classification dataset [35] that consists of 1000
classes.\"
    }
  },
  "EvaluationMetric": {

```

```

    "answer": "top-1 and top-5 error rates",
    "evidence": {
      "section": "ImageNet Classification",
      "quote": "We evaluate both top-1 and top-5 error
rates."
    }
  }
}

```

Figure A5: Full extraction output for ResNet [D4].

### A.1 Gold labels (six papers)

| Paper                   | Gold TechnicalMethod                                       | Gold Task                        | Gold Dataset                            | Gold Evaluati onMetri c   |
|-------------------------|------------------------------------------------------------|----------------------------------|-----------------------------------------|---------------------------|
| Transfo<br>rmer<br>[D6] | "Transformer"                                              | "machin<br>e<br>translati<br>on" | "WMT"                                   | "BLEU"                    |
| BERT<br>[D3]            | "BERT"                                                     | "GLUE"<br>or<br>"SQuAD<br>"      | "BooksCor<br>pus" or<br>"Wikipedia<br>" | "F1" or<br>"accurac<br>y" |
| AlexNet<br>[D5]         | "convolutional" (pa<br>per predates the<br>name "AlexNet") | "object<br>recogniti<br>on"      | "ImageNet<br>"                          | "top-1" or<br>"top-5"     |
| ResNet<br>[D4]          | "residual"                                                 | "image<br>recogniti<br>on"       | "ImageNet<br>"                          | "top-1"                   |
| MapRe<br>duce<br>[D2]   | "MapReduce"                                                | "distribu<br>ted"                | "TeraSort"                              | "seconds<br>"             |
|                         | "PageRank"                                                 |                                  |                                         | "quality"                 |



| <b>Paper</b>             | <b>Gold<br/>TechnicalMethod</b> | <b>Gold<br/>Task</b> | <b>Gold<br/>Dataset</b> | <b>Gold<br/>Evaluati<br/>onMetri<br/>c</b> |
|--------------------------|---------------------------------|----------------------|-------------------------|--------------------------------------------|
| Google<br>Search<br>[D1] |                                 | "web<br>search"      | "million<br>pages"      |                                            |

Table A1: Gold labels used for evaluation (six papers).

## A.2 proto2 background data

| <b>Paper</b>          | <b>TM</b> | <b>Tas<br/>k</b> | <b>Datase<br/>t</b> | <b>EM</b> | <b>Hypothesis<br/>set</b> |
|-----------------------|-----------|------------------|---------------------|-----------|---------------------------|
| Transformer [D6]      | 14        | 0                | 0                   | 16<br>0   | verbose_v1                |
| BERT [D3]             | 62        | 23               | 15                  | 13        | short                     |
| AlexNet [D5]          | 51        | 6                | 11                  | 4         | short                     |
| ResNet [D4]           | 51        | 6                | 14                  | 12        | short                     |
| MapReduce [D2]        | 15<br>1   | 24               | 3                   | 5         | short                     |
| Google Search<br>[D1] | 69        | 21               | 8                   | 29        | short                     |

Table A2: proto2 sentence-count output per role (six papers). See Table A3 for the per-paper pass/fail breakdown.

| <b>Paper</b>    | <b>TechnicalM<br/>ethod</b> | <b>Tas<br/>k</b> | <b>Data<br/>set</b> | <b>Evaluation<br/>Metric</b> | <b>Total</b> |
|-----------------|-----------------------------|------------------|---------------------|------------------------------|--------------|
| Transform<br>er | ○                           | ✗                | ○                   | ✗                            | 2/4          |
| BERT            | ○                           | ○                | ○                   | ○                            | 4/4          |
| AlexNet         | ○                           | ○                | ○                   | ○                            | 4/4          |

| Paper         | TechnicalMethod | Task | Dataset | Evaluation Metric | Total                  |
|---------------|-----------------|------|---------|-------------------|------------------------|
| ResNet        | ○               | ✗    | ○       | ○                 | 3/4                    |
| MapReduce     | ○               | ✗    | ✗       | ○                 | 2/4                    |
| Google Search | ✗               | ○    | ○       | ○                 | 3/4                    |
| <b>Total</b>  |                 |      |         |                   | <b>18/24<br/>(75%)</b> |

Table A3: proto2 extended gold-label evaluation results (substring match, six papers). ML papers (Transformer, BERT, AlexNet, ResNet) scored 13/16 (81%); systems papers (MapReduce, Google Search) scored 5/8 (63%). ResNet scored ✗ on Task because "image recognition" does not appear in the 6 accepted Task sentences, likely because the paper frames the task as a competition result rather than an explicit label. MapReduce scored ✗ on Task and Dataset because "distributed" and "TeraSort" are absent from accepted sentences, consistent with the lack of standard ML benchmark structure. Google Search scored ✗ on TechnicalMethod because "PageRank" does not appear in any of the 69 accepted TechnicalMethod sentences, suggesting the algorithm name is mentioned in sentences classified as other roles.

The manual review is summarised in Section 5.4 (Table 12). Variant B and the Task pilot outputs are in proto3/results\_b/ and proto3/results\_\*/ in the repository.

## Appendix B — PDF to TEI XML Conversion with GROBID

The conversion is a preprocessing step outside the notebook pipeline, which takes TEI XML as its input. I run GROBID 0.8.1 [11] locally in Docker, because a public GROBID server on HuggingFace failed in my test. The server starts with `docker run -d --rm -p 8070:8070 --name grobid lfoppiano/grobid:0.8.1`, and it is ready when `http://`

localhost:8070/api/isalive responds. A short Python script posts each PDF to `http://localhost:8070/api/processFulltextDocument` (timeout 120 s) and saves the response as a TEI XML file. `docker stop grobid` stops the server. GROBID keeps the reference list outside the body and assigns section headings, which Stage 0 needs.